

Motion2Scene: Constructing Training Scenes from Executed Humanoid Motion Contrasts

Anonymous submission

Abstract—Scenes compatible with a motion need not teach a robot when to change its behavior. We investigate training-scene construction for a humanoid selector choosing between walking and a fixed crouching command under a frozen controller. Motion2Scene uses executed command alternatives to propose overhead constraints and paired closed-loop simulation to label both command outcomes. We compare uniform, target-only, analytic contrast, and learned contrast construction. Under a nominal-placement contract, each arm supplies fifteen labeled encounter groups; analytic and learned contrast construction each acquire nine useful contrasts, while the other arms acquire none. Their selectors pass 24/36 and 22/36 evaluation conditions, versus 14/36 for either non-contrast construction. A post-hoc matched-command audit shows that always requesting the adaptation also passes 24/36. Analytic construction matches this passage while reducing unnecessary requests from 14/14 to 8/14 both-pass cases; learned construction makes 6/14 unnecessary requests but misses two useful adaptations. A retained perturbation-contract failure exposes scarce geometric eligibility. Full-observation collisions with conflicting labels explain an irreducible fitting-loss component in the learned corpus. The results support executed-contrast targeting at this label budget on a finite, partly observed development bank, without establishing learned-generator superiority, equivalence, or source-held-out transfer.

I. Introduction

A humanoid that can walk and crouch must still decide when crouching is useful. Collecting motions that demonstrate these capabilities does not directly provide training data for that decision. An overhead beam may permit both commands, defeat both, or permit only one. Moreover, a beam that clears a crouching reference may contact the robot while its controller enters the crouch. A scene constructor must therefore reason about the command the robot executes from its current history, not only the intended pose.

We ask: Which constructed scenes teach a frozen humanoid controller’s selector when to adapt? Our controlled task has one beam family and one decision: commit to walking or request an existing crouching reference, d040, at 0.30 s. The low-level SONIC controller and its command interface remain fixed. Recorded empty-scene executions provide geometric forecasts; paired obstacle-present executions determine the labels. A common learner predicts the two physical outcomes from ideal rays and robot state. Figure 1 illustrates the distinction between the reference and executed command.

This design separates four questions that are easily conflated: whether a geometric contrast exists, whether it survives execution, whether the selector can distin-

guish its observations, and whether the resulting corpus improves decisions. We vary construction while fixing the downstream learner. The analytic constructor is a strong comparator within the framework. The learned proposal is an ablation whose value is measured, rather than an assumed source of superiority.

Our contributions are (i) a command-bound scene-construction and paired-labeling protocol retaining all four physical outcomes; (ii) a finite-bank comparison showing the utility and limits of contrast-targeted training at fifteen groups per arm, including simple policies; and (iii) diagnostics of perturbation eligibility and complete-input label aliasing. The perturbation study and the nominal study remain distinct. The nominal revision followed inspection of earlier evaluation outcomes; its inference is explicitly conditional on this development bank. Neither completed simulations nor small descriptive p -values imply deployment robustness or a new-motion-source result.

II. Related Work

Learning from Hallucination (LfH) constructs constrained environments from motion collected in open space [1]. Learning from Learned Hallucination (LFLH) learns obstacle distributions with an encoder and a planning-based decoder, then trains reactive planners on rendered observations [2]. LfH-CP factorizes dynamic-scene generation into critical-point identification and obstacle trajectories passing through those points, and measures downstream navigation performance [3]. Thus, neither generating environments from motion nor evaluating their learning utility is new by itself. Our distinction is the combination of paired executed humanoid command alternatives, physical verification under a frozen controller, and a controlled study of which resulting labels support a fixed adaptation decision. We do not assert that the paired-label representation is necessary; construction also changes positive-class coverage and other corpus properties.

HumanoidPF learns whole-body traversal using a humanoid-obstacle potential-field representation and hybrid scene generation [4]. Perceptive Humanoid Parkour composes motion-matched skills and trains a perceptive policy for long-horizon traversal [5]. These systems address broader capability and deployment questions. Our idealized sensing and two-command task instead isolate training-data construction; their system-level performance is not a baseline for our finite bank. SONIC

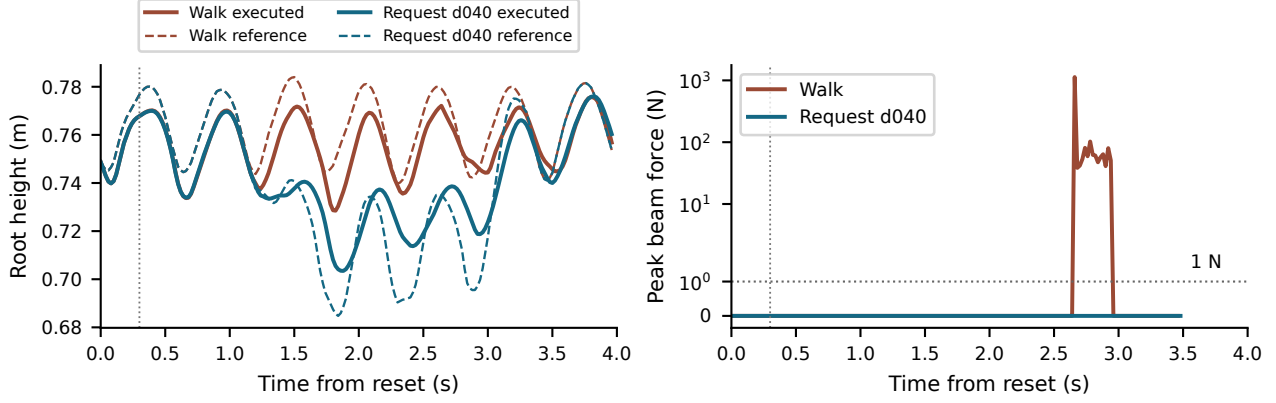


Fig. 1. Why condition construction on execution? One nominal analytic training encounter on carrier C1, label seed 8722. Left: commanded reference-bank root height (dashed) and executed height (solid), with the fixed 0.30s decision marked. Right: recorded per-body peak beam force through each command’s scoring horizon. Walking contacts the beam; d040 passes. Root height illustrates tracking differences and is not itself a clearance certificate. The method uses complete achieved body trajectories for geometry and paired physical outcomes for labels.

supplies the inherited tracking capability [6]. We neither retrain it nor attribute its locomotion skill to Motion2Scene.

Evaluation uncertainty depends on the number and structure of experimental units [7]. Matched comparisons do not remove dependence among pairs sharing a carrier or layout [8]. We retain the registered condition-level analysis, report carrier and layout structure, and distinguish a failure to detect a difference from equivalence [9].

III. Command-Bound Construction and Learning

A. Task, commands, and physical outcomes

Let $S(q)$ be a static beam scene, x_t the measured robot state, h_t its approach history, and $a \in \{0, 1\}$ a supported command. The outcome is $Y_a = Y(S(q), x_t, h_t, a; \xi)$, conditional on controller and physics seed ξ . Command $a = 0$ commits to walking. Command $a = 1$ requests d040 at 0.30s and returns to walking through the existing 3.3–3.5s legality window. This is one scene-dependent decision. Subsequent return checks use phase and reference continuity, not a new policy decision. Requests can be denied by the shared interface and are reported separately from successful adaptations.

The frozen endpoint is contact-qualified passage in the first episode. All recorded body origins must pass at least 0.10m beyond the beam’s downstream face and remain across that boundary and upright for 0.30s. Upright means root height at least 0.50m and the vertical component of the projected upward direction at least 0.50. No per-body beam force may exceed 1 N before the qualifying interval finishes. At 50Hz, the stability window includes sixteen samples. Direct 200Hz forces are aggregated by the maximum-norm substep for each body, with synchronization checked against the recorder.

The execution driver requests one 200-control-step reference pass; the records contain 199 samples over 3.96s when no reset occurs. The first reset truncates

the scored episode. Without a qualifying interval, force is evaluated through that episode’s end. Later recovery cannot rescue an earlier failure. Falls, resets, denied requests, entry/return, and the separate reference-tracking verdict remain reported diagnostics. The endpoint concerns recorded body origins and contact samples; it is not native-collider clearance or a continuous-time collision guarantee.

An encounter group contains one scene/carrier/label-seed assignment, its decision-time observation, and both command outcomes. Its label is (Y_0, Y_1) . Both-pass groups teach walking preference; $(0, 1)$ groups provide useful adaptations; $(1, 0)$ groups warn against adapting; and both-fail groups describe command-bank inadequacy. Missing commands are unknown, not negative labels. Measurement admission requires matched recorded pre-decision state, actions, motion tokens, reference history, and complete decision inputs.

B. Geometry is a proposal criterion

The beam is 0.10m deep, 1.20m wide, and 0.10m thick. Its parameters are source-route station $s \in [0.10, 0.90]$ and underside height $z \in [1.10, 1.45]$ m. Empty-scene walk and d040 executions supply sampled achieved body geometry, including the command transition. The reference route places the beam in world coordinates. Let $c_a(q)$ be the minimum signed capsule-beam clearance along the recorded command, with positive values indicating separation. The nominal geometric contrast set is

$$\mathcal{C} = \{q : c_1(q) \geq m, c_0(q) \leq -m\}, \quad m = 0.010 \text{ m}. \quad (1)$$

These sampled capsule forecasts are distinct from Y_a . Contacts alter execution, and a geometric margin is not a physical outcome.

The original acceptance contract requires these inequalities at every member of a fixed 113-offset set

in world x, y, z , and yaw. Its extrema are ± 20 mm horizontally, ± 10 mm vertically, and ± 0.02 rad. It is a worst-case finite-set test, not a probabilistic or continuous uncertainty certificate. The nominal contract evaluates acceptance only at zero offset. Both retain the original 17-offset search objective; the 17- and 113-offset outcomes are recorded as diagnostics in the nominal study.

For intuition only, suppose a fixed horizontal pose admits one closed interval of useful heights $[L, U]$. Requiring every vertical perturbation in $[-\epsilon, \epsilon]$ to remain useful gives

$$[L + \epsilon, U - \epsilon], \quad w = \max(0, U - L - 2\epsilon). \quad (2)$$

The interval is empty when the lower endpoint exceeds the upper; equality leaves a singleton of zero length. This conditional one-dimensional erosion does not establish interval connectivity, endpoint geometry, or robustness for our full pose-space test. In particular, our measured 20–27 mm best joint margins are not interval widths.

C. Four frozen construction arms

All arms receive the same carrier references and use the same scene domain. Uniform draws sixteen outputs per carrier without a future-contrast filter. Analytic construction uses achieved-motion envelopes, a global 136-candidate search initialized over twenty stations, and distinct-output ranking. Motion2Scene uses a frozen conditional eight-component proposal model and bounded pattern search. Its temporal encoder consumes a 64-frame normalized target-reference sequence and predicts mixture weights and correlated two-dimensional latent Gaussians. Sigmoid transforms map samples to eight station intervals and the common height domain. The inherited model is used with the exact d040-bound input and is not refitted on the evaluation bank.

Each learned output receives seventeen candidate queries: the initial point and four coordinate directions at four decreasing scales. Search stays within the original ± 0.05 station and ± 0.03 m height box. Target-only uses the separately frozen no-contrast checkpoint and target-only search and acceptance; it cannot regain a walking-interference objective through ranking. This arm is a construction ablation, not a same-corpus single-label control.

All arms reject nonfinite/out-of-domain outputs, within-arm duplicates, reserved layout neighborhoods, and inadequate pre-decision clearance. The contrast arms add both margin constraints; target-only adds target clearance. Visibility is measured but never gates eligibility or selection. Every output and rejected assignment is retained. Selection precedes physical labeling; no positive-only refill follows the outcomes.

D. One fixed selector

The selector receives 214 values: 144 upper/conditional-lower ray features and 70 phase,

active-skill, observation-age, gravity, root-velocity, and joint-state values. Hit distance and hit height are geometric observations; object paths, source IDs, and true beam parameters are excluded from the feature function. Robot state can nevertheless encode carrier identity indirectly. Ideal rays do not constitute a realistic perception-transfer test.

Two logistic heads predict command feasibility. For G complete groups,

$$\mathcal{L} = \frac{1}{2G} \sum_{g=1}^G \sum_{a=0}^1 \text{BCE}(p_{ga}, Y_{ga}) + 0.01 \frac{\|W\|_F^2}{428}. \quad (3)$$

Input scales are one except the 29 joint velocities, scaled by five. Parameters start at zero and receive 2,000 full-batch Adam updates at 0.01. Reported BCE excludes the weight penalty. There are no class weights or test-selected checkpoints. The linear function is compiled into the existing execution container and checked for matching probabilities and actions.

At threshold 0.5 the selector prefers walking if its walk head is positive; otherwise it requests d040 if that head is positive. If neither is positive, it records a refusal and continues walking. Refusal is a classification output, not a protective stop. Successful adaptation means a requested, executed d040 command that achieves the passage endpoint; the number of requests alone does not measure successful adaptation.

IV. Protocols, Provenance, and Metrics

Three previously inspected development carriers, denoted C1–C3 (source IDs 41001–41003), share the command qualification procedure. Seed 8721 supplies empty-scene proposal geometry; seed 8722 supplies paired labels. The fixed evaluation bank crosses stations $\{.35, .45, .55, .65\}$ with underside heights $\{1.18, 1.27, 1.36\}$ m. A condition is a carrier, layout, and physics seed. It is not an independent motion source.

Table I resolves the 24-versus-15 and 72-versus-36 denominators. The six shared groups are C1 absent/raised, C2 absent/blocked, and C3 raised/blocked, all at station .60. Each type occurs twice. Shared pairs are acquired once, even though they enter every arm’s fitting set. The nominal record retains obsolete metadata fields naming six generated slots per carrier and 72 requested groups; its operative three-eligible quota, code, and executed assignments agree on 36 generated groups and 72 new commands. These metadata discrepancies do not justify changing the historical files.

The nominal protocol was recorded after a 366-execution parent snapshot was available: 61 parent conditions, including 31 of the 36 nominal conditions, had already been evaluated. The earlier learner revision also used inspected layout outcomes. Nominal proposals and labels were frozen before their own execution, but this is not an untouched confirmatory test. The panels reuse all twelve layouts, three carriers, one evaluation seed, six

TABLE I

Protocol and deviation summary. Both studies retain proposal seed 8841, label seed 8722, controller, commands, learner, and scorer.

Item	Perturbation study	Nominal study
Acceptance	113 fixed offsets; contrast margin 10 mm	Zero-offset acceptance; same margin; perturbations diagnostic
Selection	First six outputs per carrier/arm assigned before audit	First three eligible outputs per carrier/arm selected from sixteen
Groups per arm	18 requested generated + 6 shared backgrounds = 24	9 selected generated + the same 6 backgrounds = 15
Realized groups	Uniform 24; analytic 7; target-only 22; learned 6	15 in every arm; no refill or exclusion after labels
Evaluation	12 layouts $\times$ 2 seeds $\times$ 3 carriers = 72 conditions; six policies	Seed 8511 only: 36 conditions; four new policies; two fixed comparators reused
Backgrounds	108 separate executions: three types, two seeds, three carriers, six policies	Original background labels reused for fitting; parent-model control results reported separately
Inference	Unequal acquired data at equal requested slots	Equal group count; outcome-informed revision on overlapping development data

background labels, the generator checkpoints, and the controller. Nineteen selected nominal generated scenes also match previously labeled parent scenes at the same label seed (nine uniform, nine target-only, one analytic). They were executed again under the nominal protocol. The two panels are not independent replications.

The registered primary estimand is mean passage within each carrier, averaged over the three carriers; balanced denominators make it numerically equal to the pooled fraction. We target finite-bank performance, conditional on the selected training corpora. New-layout population performance and new-source generalization would require different sampling and holdout designs.

Registered comparisons pair conditions and report every cell of the 2×2 table. Exact two-sided binomial tests condition on discordances. Their independence assumption is not established across reused layouts or carriers; we retain them as descriptive calculations, not population uncertainty. Physics repeats stay with their carrier/layout. Deterministic fitting removes optimizer randomness, not training-corpus variability.

V. Results and Diagnostics

A. Acquisition at fifteen groups per arm

All 72 nominal label commands and 144 nominal policy executions pass measurement admission. The four primary fits replay with exactly matching weights, biases, scales, and BCE. Table II separates output count, geometric eligibility, selected groups, and useful physical contrast. Analytic and learned construction each provide nine useful contrasts. Uniform and target-only provide none. Thus, nine useful contrasts out of nine selected generated groups does not mean nine proposals or equal acquisition computation.

All six nominal predictions hold: each contrast arm supplies at least one eligible group on each carrier (three selected each); an arm other than analytic acquires useful contrasts (learned, nine); uniform yields at most one (zero); target-only is predominantly both-pass (9/9

generated, 13/15 including backgrounds); precisely the arms with useful labels request d040 at evaluation; and most nominally eligible outputs fail the 113-offset contrast diagnostic (167/168). The last count includes untargeted arms; among selected contrast-arm outputs it is 17/18. No visibility filtering is applied.

B. Passage and the always-adapt alternative

Analytic and learned construction pass 24/36 (66.7%) and 22/36 (61.1%) conditions, versus 14/36 (38.9%) for either non-contrast construction (Table III). Their improvements are 27.8 and 22.2 percentage points on this bank. Both directions hold on all three carriers. In C1–C3 order, counts are 4/12, 5/12, 5/12 for uniform and target-only; 8/12 on every carrier for analytic; and 7/12, 7/12, 8/12 for learned construction. Figure 2 preserves layout-level outcomes.

No reverse discordances occur against uniform or target-only. Analytic exceeds learned construction by two conditions (5.6 percentage points), both in the analytic direction. No statistically significant difference was detected by the registered test ($p = 0.5$); equivalence was not established. A post-hoc carrier sign calculation has only three positive clusters against uniform and gives $p = 0.25$ if those carriers were independent. We do not use condition counts to construct narrow new-source confidence intervals.

The strongest simple comparison changes the interpretation. Scripted rays also pass 24/36, and privileged geometry passes 20/36. Eight nominal conditions lacked d040 outcomes in the combined records. A separate post-hoc protocol executed both constant commands on those conditions, retaining the original settings. All eight repeated walks reproduce passage and recorded pre-decision history exactly. Across original and audit executions, all 304 available same-command comparisons match both passage and the recorded state/action/token trajectory fields. The full 214 inputs, beam, seed, banks, non-policy configuration, and scoring contract

TABLE II

Nominal acquisition and fitted-label composition. Each arm proposes 48 outputs (16 per carrier), selects nine generated groups, and reuses six background groups. Eligibility is geometric; outcome columns are physical paired labels over all fifteen groups. Every group has two labels.

Construction	Outputs	Eligible	Groups	Both pass	d040 only	Walk only	Neither	BCE
Uniform	48	48	9 + 6	7	0	0	8	0.001762
Analytic	48	41	9 + 6	4	9	0	2	0.000542
Target-only	48	46	9 + 6	13	0	0	2	0.000175
Motion2Scene	48	33	9 + 6	4	9	0	2	0.139004

TABLE III

The same 36 nominal conditions. Requests on both-pass cases are unnecessary for passage. Both-fail requests and refusals do not avoid failure. The scripted and privileged geometric comparators reuse their registered seed-8511 executions. *Post-hoc command lookup with eight missing conditions completed by sixteen separately registered matched executions. The outcome oracle uses physical labels in hindsight; privileged geometry uses a forecast.

Policy	Passage	Requests	Both-pass req.	Both-fail req.	Refusals
Uniform	14/36	0/36	0/14	0/12	17/36
Analytic	24/36	24/36	8/14	6/12	0/36
Target-only	14/36	0/36	0/14	0/12	0/36
Motion2Scene	22/36	22/36	6/14	8/12	0/36
Scripted rays	24/36	24/36	8/14	6/12	0/36
Privileged geometry	20/36	7/36	1/14	0/12	17/36
Always walk*	14/36	0/36	0/14	0/12	0/36
Always d040*	24/36	36/36	14/14	12/12	0/36
Outcome oracle*	24/36	10/36	0/14	0/12	12/36

TABLE IV

Registered nominal paired comparisons. All four cells are shown. The condition-level p -values assume independent discordances; they are descriptive because conditions share carriers and layouts. An.: analytic; Un.: uniform; Tgt.: target-only; M2S: Motion2Scene.

A / B	Both	A only	B only	Neither	p
M2S / An.	22	0	2	12	0.5
M2S / Un.	14	8	0	14	0.0078125
An. / Un.	14	10	0	12	0.001953125
M2S / Tgt.	14	8	0	14	0.0078125

TABLE V

The original perturbation study, kept separate. Its equal-24 complete-group goal fails in three arms. These are equal requested slots with unequal acquired data, not the nominal fifteen-group comparison.

Construction	Groups	Useful	Passage
Uniform	24/24	0	28/72
Analytic	7/24	1	39/72
Target-only	22/24	0	28/72
Motion2Scene	6/24	0	28/72

also match across methods. These checks support command lookup for this one-decision deterministic runtime; they are not a hidden-state snapshot or an evaluation method for sequential policies.

The completed command table contains fourteen both-pass, ten d040-only, twelve both-fail, and zero walk-only conditions. Always-d040 therefore passes 24/36, reaching the two-command outcome oracle. Analytic matches this passage with 24 rather than 36 requests, avoiding six unnecessary adaptations on the fourteen both-pass cases. Learned construction avoids eight unnecessary adaptations, but requests d040 on only eight of ten rescue cases and passes 22/36. Its extra selectivity does not imply a superior tradeoff. The scripted rule makes exactly the same requests and obtains the same outcomes as the nominal analytic selector on all 36 conditions.

Neither contrast selector refuses on the nominal traver-

sal bank. They request d040 on 6/12 and 8/12 both-fail conditions, respectively. In the parent background suites, all six policies make zero requests over the 72 absent/raised executions. All four parent learners and privileged geometry refuse 6/6 blocked cases, whereas scripted rays refuse 0/6; all blocked executions fail. These are parent-model controls, not evaluations of the new nominal fits. Successful refusal or safe stopping is not demonstrated. The recorded nominal analytic and learned selectors each make eighteen successful d040 passages despite making 24 and 22 requests, respectively. Resets occur in 5/36 and 4/36 of their traversal runs; their nominal return counts are 21/24 and 19/22 requests. Passage and complete tracking remain different endpoints.

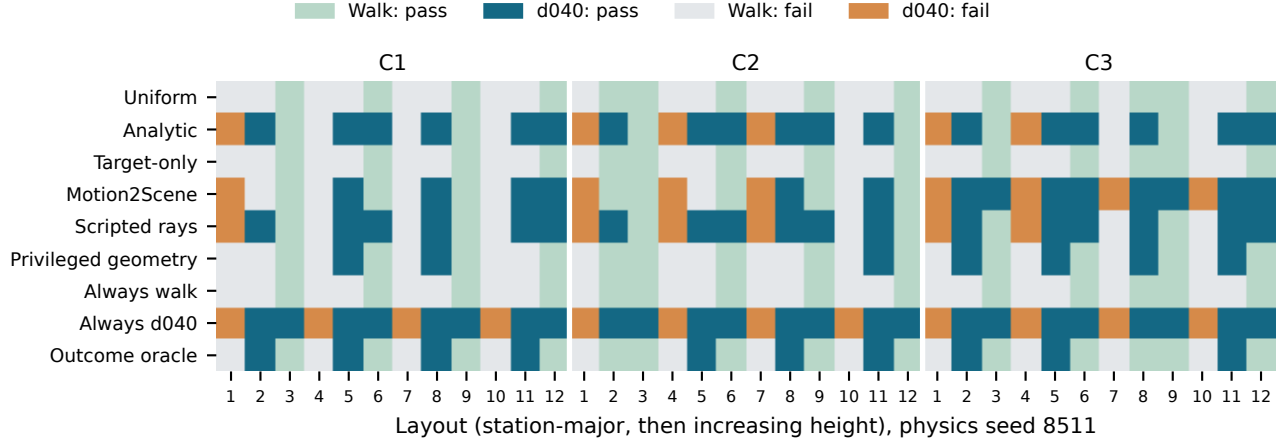


Fig. 2. Where do decisions matter? All nominal carrier/layout conditions, with the two commands’ outcomes completed by the post-hoc matched audit. Within each carrier, layouts enumerate four increasing stations, each with three increasing heights. Color reports actual command and passage; a refusal executes walk and is included in that command’s category. The learned selector’s two missed rescues remain visible, as do unnecessary adaptations and conditions neither command solves.

C. Retaining the failed perturbation contract

The original study completes 82 labeling and 540 evaluation executions: 432 traversal runs over 72 conditions with six policies, plus 108 separate background controls. Only one generated group, analytic on C1, provides a useful contrast. The learned arm receives only six shared backgrounds (Table V). Its result cannot establish the utility of learned generated scenes because none entered its corpus.

The prediction that each contrast arm supplies a useful group on every carrier fails. Uniform’s at-most-one prediction holds (0/18 requested), as does target-only’s at-least-twelve both-pass prediction (16/18). All 23 executed generated target-clear predictions achieve passage, satisfying the zero-false-clear prediction in this selected set; eight receive separate tracker rejections. The prediction of at least three net improvements for each contrast arm fails jointly: analytic adds eleven passages, learned adds zero. All failed predictions remain part of the interpretation.

For the registered paired comparisons, analytic versus uniform has (both-pass, analytic-only, uniform-only, both-fail) = (28,11,0,33). Learned versus analytic has (28,0,11,33); learned versus uniform and target-only each have (28,0,0,44). The eleven-discordance calculation gives $p = 0.0009765625$; zero-discordance comparisons contain no directional information. Physics repeats remain grouped. Scripted rays pass 46/72, privileged geometry 39/72. These parent results support the importance of acquired contrasts, but are not a replication of the nominal comparison.

The stored envelope diagnostic holds recorded execution geometry fixed (Fig. 3). Its specified 81-station by 141-height grid has 11,421 centers; the protocol’s parenthetical 7,181 total is an arithmetic error. Nominal contrast support shrinks sharply under the finite offset

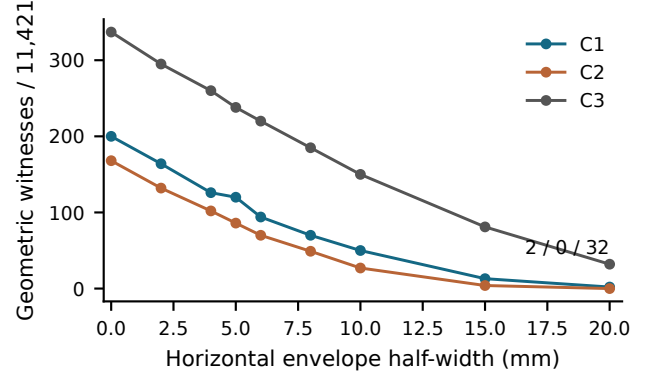


Fig. 3. How much eligibility survives the uncertainty requirement? The existing post-hoc grid diagnostic evaluates 11,421 station/height centers per carrier. Horizontal half-width is shown; vertical half-width is half that value and yaw scales to 0.02rad. Nominal witnesses 200/168/337 fall to 2/0/32 at the inherited envelope. These are finite geometric witnesses, not executed successes or continuous feasibility volumes.

test. The observed decrease does not prove continuous infeasibility. Because the set contains zero offset, excluding nominal failures from further checks is valid; zero inclusion alone does not prove monotonicity between differently scaled, non-nested finite sets. The nominal study addresses a narrower placement-known task and demonstrates no robustness at the rejected tolerance.

D. Complete-input aliasing and fitting loss

Among all nominally critical outputs, analytic has decision-time proposal ray hits in 44/47 scenes, versus 24/34 for learned construction. These denominators count geometrically critical outputs before common exclusions; they are not label counts. All nine selected analytic contrasts have ray hits, compared with five of nine selected learned contrasts. No group is removed in

this post-hoc analysis.

The fitting audit checks all 214 features, identical loss reduction, normalization, regularization, multiplicity, and class balance. Both contrast arms contain four positive walk labels and thirteen positive d040 labels over fifteen groups. Analytic BCE is 0.000541579 and learned BCE is 0.139004365, a ratio of approximately 257. Their separate weight penalties are 0.001360424 and 0.001004991. Exactly reproducing the fixed 2,000-step fits rules out an accounting mismatch; it does not by itself prove convergence.

In the learned corpus, one complete-input equivalence class contains two both-pass backgrounds and one d040-only contrast on C1. Another contains one both-pass background and three d040-only contrasts on C3. Their maximum absolute full-feature differences are zero. An arbitrary deterministic predictor must assign a common probability to each identical input. If a class has n members with positive fraction r in a head, its minimum summed BCE is $nH(r)$, where H is binary entropy. Summing over classes and heads, and dividing by thirty labels, gives an empirical infimum of 0.138629 for this corpus. The measured learned BCE is only 0.000375 higher. Analytic has no conflicting exact-input class and zero observed head classification errors; learned has two, matching its minimum possible head-error count. The post-hoc six-, four-, and two-decimal comparisons remain precision diagnostics, not substitutes for exact collisions.

This is direct evidence for an observation limitation in fitting walk feasibility, beyond an aggregate correlation with ray visibility. It does not isolate invisibility as the cause of downstream performance or of every generator difference. In particular, d040 succeeds for every member of both conflicting classes: always adapting can resolve their passage without resolving whether adaptation is necessary. Shared both-pass and both-fail examples remain useful training information.

E. Dependence on the acquired corpus

The twenty-four registered parent leave-four-assignment-out refits are retained, including missing and zero-member folds. Removing the single analytic contrast produces a model exactly equal to the background-only learned-arm model. On ninety recorded parent condition inputs, that refit’s requests fall from 29 to zero; the other folds produce 29, 29, 31, 29, and 29 requests. This recorded-input comparison is a post-hoc diagnostic, not new physical evaluation of refit policies.

We additionally refit the unchanged nominal learner after removing a carrier’s entire training set, including its backgrounds, and after removing each contrast group separately. These thirty post-hoc CPU fits use matched command lookup and never replace the primary selectors. Leaving out C1/C2/C3 yields 8/12, 6/12, 8/12 passages on the removed carrier for analytic, and 8/12, 7/12, 7/12 for learned construction. Uniform and target-only retain 4/12, 5/12, 5/12. Removing one contrast

TABLE VI

Measured rollout-process time, including startup and contention. The first four rows sum to 6.577196 h (the historical “6.6 GPU-hour” boundary). This is not GPU utilization time or total method training cost.

Study	Stage	Runs	Process hours
Robust	labels	82	0.824108
Robust	evaluation	540	4.310964
Nominal	labels	72	0.480848
Nominal	evaluation	144	0.961276
Post-hoc audit	evaluation	16	0.137223

gives analytic 21–24/36 and learned 22/36 on the full bank. These are training-set sensitivity measurements, not independent training datasets or confidence intervals. The generator still includes development-source information; selector-only removal establishes no generator-level source-held-out transfer.

F. Resource accounting

Table VI counts every executed label and evaluation in the two studies, with shared labels and reused comparators charged once. The sixteen new audit executions add 0.137223 h separately. The shared empty-transition bank costs another 0.105115 h outside that historical boundary. Controller training, motion/generator pretraining, earlier experiments, CPU search, and idle scheduling time are excluded. Original selector-fit CPU wall time was not recorded, so no complete end-to-end efficiency claim is made.

Nominal construction preparation takes 38.820 CPU wall seconds. Per-arm recorded construction/search/audit loops take 3.491, 9.420, 8.955, and 15.984 s for uniform, analytic, target-only, and learned construction. Their search-clearance query counts are 0, 13,872, 13,872, and 27,744, respectively, where one query evaluates a candidate, offset, and command. Common verification and overhead remain additional to that query counter. Equal-label comparison is therefore not equal-compute comparison. Historical fresh-source search timing concerns a different reference-only apparatus and is not evidence of nominal selector efficiency or transfer.

VI. Limitations and Conclusion

This study uses one controller, one beam family, ideal simulator rays, one adaptation decision, and three inspected development carriers. Its fixed bank was partly observed before nominal registration. Generalization to new layouts drawn from a population, new motion ancestry, realistic perception, or hardware remains untested. Finite geometry checks and contact-qualified passage provide neither robust deployment guarantees nor a protective refusal behavior. Corpus-removal refits expose some training-data dependence but do not estimate acquisition variability.

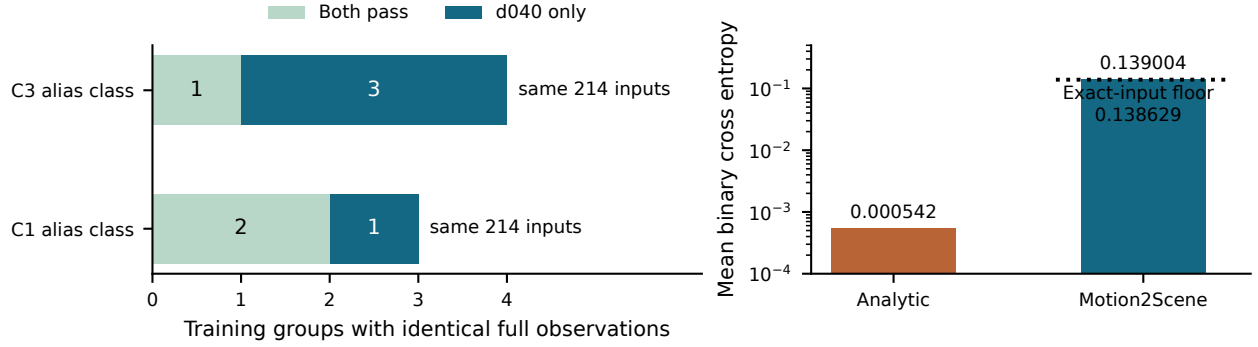


Fig. 4. Can the complete selector observation distinguish its labels? Left: two exact full-input equivalence classes in the learned corpus combine both-pass backgrounds with d040-only contrasts. All 214 values are identical within each class. Right: fitted mean BCE and the unrestricted deterministic predictor’s empirical BCE infimum from these collisions. Identical rays alone would not establish this bound. Both classes share successful d040, so label ambiguity is not an unavoidable passage error. All original groups are retained.

At fifteen groups per arm, executed-contrast targeting supplies the adaptation examples absent from uniform and target-only corpora and improves measured finite-bank passage. Always-d040 and scripted rays show the limit of that result: learned scene construction does not establish superior traversal capability. Analytic reaches the available command ceiling while avoiding some unnecessary adaptations; learned construction trades two missed rescues for fewer unnecessary requests. Preserving the failed perturbation contract and auditing complete-input aliasing make the framework’s restrictions explicit. The contribution is a measured account of constructing and verifying decision-relevant training data, not equivalence, learned-generator superiority, or a claim that paired labels are necessary.

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